

Beyond Contraction: Evidence for Sense-Directed Reorganization in Contextual Lexical Representations

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Abstract

Context often makes the intended sense of an ambiguous word easier to recover, but this does not imply that contextual representations should collapse into a tighter within-sense cluster. We test that geometric intuition in frozen BERT-base, RoBERTa-base, and DeBERTa-v3-base representations of 19 coarse ambiguous nouns. Controlled sense-diagnostic evidence increases between-sense separation relative to matched random context in every aggregate model-layer-evidence cell, while within-sense dispersion can decrease, remain nearly unchanged, or increase. A nested intervention then holds the pre-existing context fixed and adds one diagnostic or matched control cue. Diagnostic cues increase *linearly accessible* sense information and increase alignment of representation updates with a supervised sense-discriminative subspace, but they also alter broader representational structure: CKA with the pre-cue state decreases and residual variation often increases. Fresh decoders recover a fixed low-dimensional representation of the old context at similar levels to controls, whereas base-trained decoders transfer less well, a pattern consistent with reorganization rather than simple information loss. The same qualitative picture extends to label-independent local context expansion: sense accessibility usually increases as nearby context grows, and descriptive correlations between accessibility gain and supervised sense-direction alignment are positive across all nine model/layer settings. Finally, LEACE substantially reduces linearly accessible sense information while causing only a small reduction in old-context reconstruction, and contextual similarity on RAW-C tracks graded human semantic relatedness. We therefore use *sense-directed reorganization* as a descriptive summary of the observed geometry, not as a claim of a universal computational mechanism.

1 Introduction

A word such as *bank* can denote a financial institution or a river edge. Once a context contains *mortgage*, the financial interpretation becomes much easier to identify. A natural geometric intuition is that contextual evidence should progressively narrow the representation toward the selected sense,

$$R_0 \supset R_1 \supset R_2 \supset \dots \quad (1)$$

This intuition is attractive but conflates two objects. A contextual encoder produces one deterministic vector $h = E(w, c)$ for one occurrence, whereas a “cloud” is formed only by collecting representations across many contexts. The distribution of representations across contexts is therefore not itself a probability distribution over lexical senses.

Our experiments do **not** directly estimate semantic uncertainty such as $H(S | C)$. Instead, we ask how easily a coarse lexical sense can be recovered by a fixed family of linear probes and how the geometry of the representation changes as contextual evidence is altered. The key distinction is thus

$$\text{sense accessibility} \neq \text{within-sense geometric dispersion.} \quad (2)$$

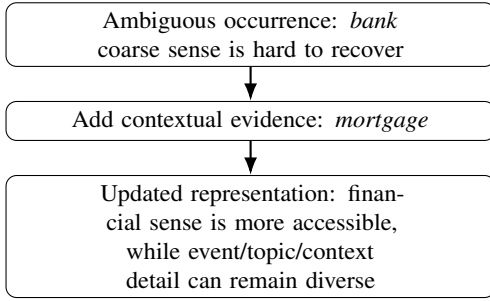
Two sentences may make the same sense obvious while describing different events, topics, participants, or syntactic structures. Increased sense accessibility therefore need not imply lower representational variability.

Prior work has established strong context sensitivity and anisotropy in contextual token representations [Ethayarajh, 2019], recoverable sense distinctions [Loureiro et al., 2021], nontrivial variation among same-sense occurrences [Wang and Zhang, 2023], and structured context-induced concept transformations [Hu et al., 2026]. We focus on the missing intersection of these observations: *how does the geometry of a fixed ambiguous lexical item change as sense-relevant evidence is supplied?*

We answer this with three complementary experiments. First, a controlled oracle intervention compares sense-diagnostic evidence with matched random context. Second, a nested intervention keeps the entire pre-existing context identical and adds one diagnostic or control cue, allowing us to distinguish sense-accessibility gain from retention of old contextual structure. Third, we test label-independent local context expansion, linear sense erasure, and external semantic alignment on RAW-C.

Our contributions are empirical and deliberately narrow:

- Across the tested encoders and coarse ambiguous nouns, controlled diagnostic evidence consistently increases between-sense differentiation, but no universal within-sense contraction law appears.
- Under a nested intervention, sense information becomes more linearly accessible and updates align more strongly with a *supervised* sense-discriminative subspace, while broader structure also changes.
- Old-context information remains largely linearly recoverable under a fixed reconstruction target, and substantial linear



Question: does the cloud contract, do senses separate, or is the representation reorganized more broadly?

Figure 1: Conceptual setup. Sense accessibility and representational dispersion are different quantities.

sense erasure causes only limited collateral context loss.

- A label-independent local-context protocol and RAW-C provide complementary robustness and external semantic-validity checks.

2 Related Work

Contextual geometry and lexical ambiguity. Contextual representations vary substantially with context and are anisotropic [Ethayarajh, 2019]. Polysemous-word geometry is not always well approximated by isolated clusters [Yenicelek et al., 2020], yet coarse senses are recoverable in contextual encoders [Loureiro et al., 2021]. Same-sense representations also retain systematic contextual variance [Wang and Zhang, 2023]. We study how these quantities change when the available evidence itself is manipulated.

Context and lexical information. Pimentel et al. [2020] show that lexical ambiguity and contextual informativeness are closely related. Their information-theoretic framing motivates our separation between sense-related predictability and geometric dispersion, but we do not equate our linear-probe quantity with Shannon mutual information or semantic entropy.

Dynamic geometry. Recent work represents concepts as point clouds transformed by context [Hu et al., 2026]; layer-wise trajectory analyses likewise study how ambiguous-token representations bifurcate through depth [Pandey and Singh, 2026]. Our primary intervention axis is contextual evidence rather than layer depth, although we retain layers 4, 8, and 12 as a secondary axis.

Probing, similarity, and erasure. Probe performance is not equivalent to causal use [Hewitt and Liang, 2019]. We therefore describe our probe results as *linear accessibility*, use CKA [Kornblith et al., 2019] and RSA as structural diagnostics, and use LEACE [Belrose et al., 2023] as a linear erasure intervention. Amnesic probing motivates the distinction between decodability and causal intervention [Elazar et al., 2021].

Human graded semantics. RAW-C contains graded human relatedness judgments for ambiguous words in natural contexts [Trott and Bergen, 2021]. We use it only as an external semantic-validity check; it does not reproduce the controlled incremental intervention and does not establish human-model mechanistic equivalence.

3 Formalization

Let S be the coarse sense label of target word w , C a context state, and $Z_\ell = f_\ell(w, C)$ the target-token representation at layer ℓ .

Linear sense accessibility. For a linear probe family $\mathcal{V}$, we report

$$I_{\mathcal{V}}(S; Z) = \text{NLL}_{\text{null}} - \text{NLL}_{\text{probe}}. \quad (3)$$

This is a probe-family-dependent reduction in predictive loss, measured in nats. It is **not** Shannon mutual information and is not used as a direct estimator of semantic uncertainty.

Covariance geometry. For sense means μ_s and global mean μ , we decompose total covariance into within- and between-sense components,

$$W = \sum_s p_s \text{tr Cov}(Z \mid S = s), \quad (4)$$

$$B = \sum_s p_s \|\mu_s - \mu\|_2^2, \quad (5)$$

$$\text{tr Cov}(Z) = W + B. \quad (6)$$

We also use the normalized sense ratio

$$R_S = \frac{B}{W + B}. \quad (7)$$

Raw magnitudes are not compared directly across model families because representation scale and anisotropy differ.

Supervised sense subspace. From training representations we define

$$U_S = \text{span}\{\mu_s - \mu\}. \quad (8)$$

For a paired update $\delta = h_{\text{post}} - h_{\text{pre}}$, the fraction

$$r_S(\delta) = \frac{\|P_{U_S} \delta\|_2^2}{\|\delta\|_2^2} \quad (9)$$

measures alignment with a *supervised* sense-discriminative subspace. Because both U_S and sense-accessibility probes use gold sense labels during training, their association is treated as descriptive evidence of sense-structured change, not as an unsupervised discovery or a proof of a unique mechanism.

4 Experimental Setup

4.1 Data and Encoders

We use CoarseWSD-20 [Loureiro et al., 2021], a Wikipedia-derived benchmark with 20 ambiguous English nouns and 2–5 coarse senses. After balancing within each lexical item and applying context-availability constraints, 19 words remain; *deck* falls below the minimum held-out examples per sense. The statistical unit for CoarseWSD analyses is the *lexical item*, not token pairs or pairwise distances.

We analyze frozen BERT-base-uncased [Devlin et al., 2019], RoBERTa-base [Liu et al., 2019], and DeBERTa-v3-base [He et al., 2021]. We extract layers 4, 8, and 12 and average multiple target WordPieces/subwords. Geometry is examined in raw and standardized coordinates; the nested experiment additionally includes a PCA-whitened diagnostic fitted only on pooled training representations and then held fixed across conditions. Because cross-model metric scales differ, our cross-architecture conclusions rely on signs and within-model trajectories rather than raw magnitude comparisons [Timkey and van Schijndel, 2021].

4.2 Experiment A: Controlled Diagnostic Evidence

Token–sense PMI is estimated from the official training split only. For each held-out occurrence, an oracle diagnostic ordering reveals tokens with highest PMI for the gold sense, while five random reveal orders provide matched context-quantity controls. We reveal $k \in \{1, 2, 4, 8, 16\}$ tokens. The oracle protocol is a controlled intervention, not a deployable WSD system. BERT additionally receives a deletion control; this artifact check is therefore BERT-specific and does not by itself rule out masking effects for RoBERTa or DeBERTa-v3.

4.3 Experiment B: Nested Context Update

Each occurrence first receives the same label-independent random base context with $m \in \{2, 4, 8\}$ visible tokens. From that identical state we construct

$$h_{\text{base}} = E(w, C_{\text{base}}), \quad (10)$$

$$h_{\text{diag}} = E(w, C_{\text{base}} \cup \{c_{\text{diag}}\}), \quad (11)$$

$$h_{\text{ctrl}} = E(w, C_{\text{base}} \cup \{c_{\text{ctrl}}\}), \quad (12)$$

where c_{diag} is the highest training-PMI remaining token and each control is matched by side and approximate distance to the target. Three control seeds are averaged.

Old-context reconstruction. The pre-existing base context is converted to a fixed lexical target with a training-only TF-IDF vectorizer (lowercase, unigram+bigrams, minimum document frequency 2, maximum vocabulary 12,000), followed by TruncatedSVD to at most 64 dimensions and row ℓ_2 normalization. A ridge decoder with fixed $\alpha = 10$ predicts this target from the contextual token representation. We report (i) a fresh decoder trained separately in each condition and (ii)

transfer of a decoder trained on the base state without refitting. TF-IDF/SVD and decoder fitting use training examples only; evaluation is on the official held-out split. Hyperparameters were fixed rather than tuned on test performance. We interpret reconstruction *comparatively*: the 64-dimensional lexical target is not claimed to capture all contextual information.

We additionally measure CKA, RSA, covariance geometry, $r_S(\delta)$, and within-sense residual variation parallel/orthogonal to U_S .

4.4 Experiment C: Validation

Label-independent local context expansion. To remove gold-sense cue selection, context positions are ordered only by distance to the target, with deterministic tie breaking. The nearest 2, 4, 8, and 16 tokens form nested context states. Gold labels are still used *afterward* to train the sense probe and supervised sense subspace; therefore “label-independent” refers only to cue selection.

Linear sense erasure. LEACE is fit on training diagnostic states and applied to held-out diagnostic states. We remeasure linear sense accessibility and old-context reconstruction with both a freshly fitted decoder and a pre-erasure decoder. LEACE supports claims about linearly accessible information only.

RAW-C. For each of 672 RAW-C sentence pairs, we compute target-token cosine similarity and compare it with graded human relatedness and same/different-sense labels.

4.5 Statistical Scope

For paired CoarseWSD effects, we bootstrap lexical items with 5,000 resamples and use paired word-level Wilcoxon tests; repeated token pairs are never treated as independent observations. Aggregate sign counts such as “27/27” or “45/45” are reported only as descriptive robustness summaries, not as independent replication counts.

The local-context correlation analysis contains three transitions for each of 19 words. Because these 57 points are clustered within lexical item, the ordinary Spearman p -values produced by the exploratory notebook are anti-conservative for paper-level inference. In this revision we therefore report *correlation effect sizes only* and make no significance claim from those p -values. A venue-submission version should replace them with a word-cluster bootstrap or permutation test and apply multiplicity control across model/layer settings. For the same reason, RAW-C inferential p -values over 672 pairs are not used as evidence here; we report only descriptive effect sizes because pairs are grouped by lexical item.

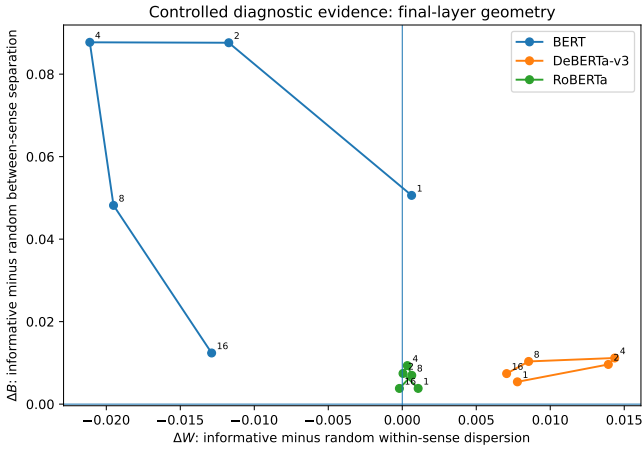


Figure 2: Controlled evidence at layer 12. Raw diagnostic-minus-random geometry is shown only within each model; magnitudes should not be compared as if the model spaces shared a common unit.

5 Results

5.1 Controlled Evidence Increases Between-Sense Differentiation

In Experiment A, the diagnostic-minus-random change in between-sense centroid separation is positive in every one of the 45 aggregate model-layer-evidence cells. We use this as a descriptive robustness statement, not as $n = 45$ independent evidence. The within-sense change is qualitatively different across encoders (Figure 2). At layer 12 and $k = 2$, for example, BERT has $(\Delta W, \Delta B) = (-0.0117, 0.0876)$, RoBERTa $(0.00005, 0.0074)$, and DeBERTa-v3 $(0.0139, 0.0096)$. Thus the common qualitative effect is stronger sense differentiation, whereas within-sense contraction is not universal.

The BERT mask/delete comparison preserves the same qualitative direction for between-sense differentiation, but this artifact control is not available for the other two encoders and therefore only partially addresses masking concerns.

5.2 Nested Updates Are Broader Than a Sense-Only Shift

Experiment B shows that the diagnostic cue increases linearly accessible sense information relative to matched controls across the aggregate grid (mean $+0.265$ nats). The update also places a larger fraction of its energy inside the supervised sense subspace (mean diagnostic-minus-control $+0.032$). At the same time, CKA with the pre-cue base state is lower under the diagnostic cue than under controls (mean -0.050), and orthogonal residual variance is positive in most aggregate cells. Figure 3 summarizes the first two effects.

A fresh decoder trained after the update changes old-context reconstruction only slightly relative to the matched control (mean diagnostic-minus-control cosine $+0.00194$), while a decoder trained on the base state transfers less well (mean

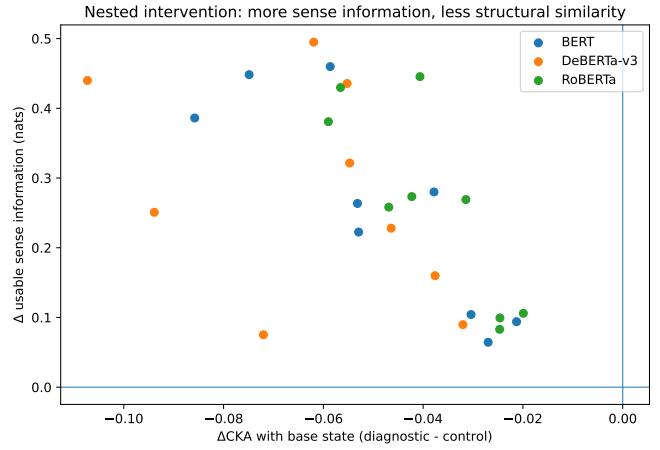


Figure 3: Nested intervention. Diagnostic cues tend to increase linear sense accessibility while reducing CKA with the identical pre-cue state relative to matched controls. The 27 aggregate points share words/models/layers and are descriptive, not independent observations.

-0.00802). Together with lower CKA, this pattern is *consistent* with a representational remapping/reorganization rather than simple loss of all old-context accessibility. It does not uniquely identify the transformation, so we avoid treating “remapping” as a proved mechanism. RSA was also computed as a structural diagnostic; its role is supplementary rather than a basis for the headline claim.

5.3 Label-Independent Local Context Raises Sense Accessibility

When cue selection is based only on local distance, linear sense accessibility usually increases as context expands. Across individual $2 \rightarrow 4$, $4 \rightarrow 8$, and $8 \rightarrow 16$ transitions, the bootstrap summaries in the executed notebook show positive mean usable-information gains for every model/layer/transition; the positive-word fraction is typically high and the gain diminishes at richer contexts.

The descriptive association between usable-information gain and supervised sense-subspace update alignment is positive in all nine model/layer settings: BERT $\rho = .629/.698/.718$ at layers 4/8/12, RoBERTa $.602/.485/.498$, and DeBERTa-v3 $.420/.304/.174$. Because each word contributes three transitions, these are effect-size summaries rather than inferential tests in the present paper.

5.4 Linear Sense Erasure Has Limited Contextual Collateral Damage

LEACE reduces linear sense accessibility substantially on held-out diagnostic states. Table 1 reports absolute pre/post quantities averaged over base-context sizes. For BERT layer 12, for example, usable information falls from 0.621 to 0.199 nats and balanced accuracy from 0.894 to 0.590, while old-context reconstruction cosine changes from 0.180 to 0.178. The pattern

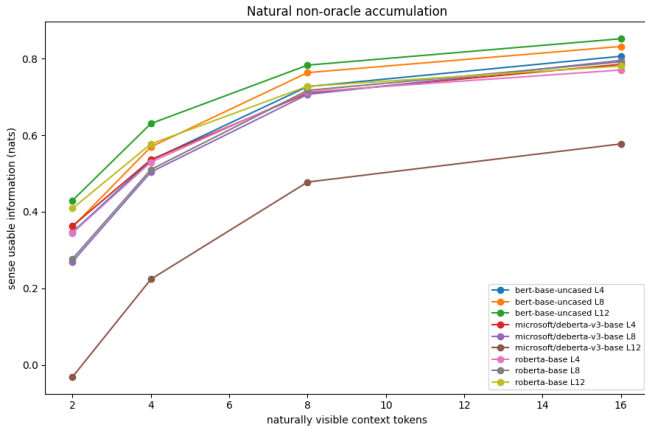


Figure 4: Label-independent local context expansion. Cue order uses distance only; sense labels are used afterward for the linear probe.

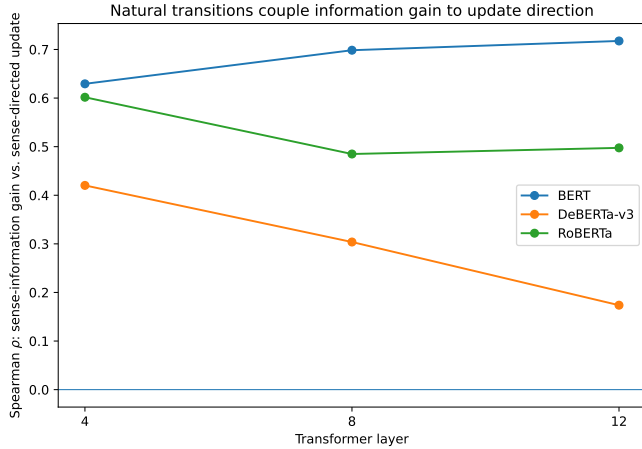


Figure 5: Descriptive Spearman association between linear sense-accessibility gain and alignment with a supervised sense-discriminative subspace under label-independent cue selection. Transitions are clustered within words; ordinary pair-level p -values are not used for inference.

is qualitatively similar across the tested model/layer settings, although DeBERTa-v3 layer 12 begins from much weaker linear sense accessibility.

The corresponding old-context reconstruction cosine decreases by roughly 1.0–1.5% relative to its pre-erasure value (Figure 6). Word-level bootstrap intervals for the per-base-size changes are included in the repository. The result supports *partial linear separability*: a large amount of linearly accessible sense signal can be suppressed without proportionate destruction of the fixed old-context target. It does not imply complete independence or the absence of nonlinear sense information.

5.5 External Semantic Alignment on RAW-C

On RAW-C, target-token cosine similarity has positive descriptive correlation with graded human relatedness in every model/layer setting ($\rho = .438\text{--}.693$), and same-versus-

Model	L	I_V pre	I_V post	Acc. pre	Acc. post
BERT	8	.587	.183	.886	.586
BERT	12	.621	.199	.894	.590
RoBERTa	8	.437	.100	.837	.563
RoBERTa	12	.497	.143	.857	.575
DeBERTa	8	.341	-.005	.815	.550
DeBERTa	12	.076	-.183	.739	.510

Table 1: Held-out linear sense accessibility before and after LEACE, averaged over base sizes. Negative I_V means the fitted probe is worse than the null predictor under this definition.

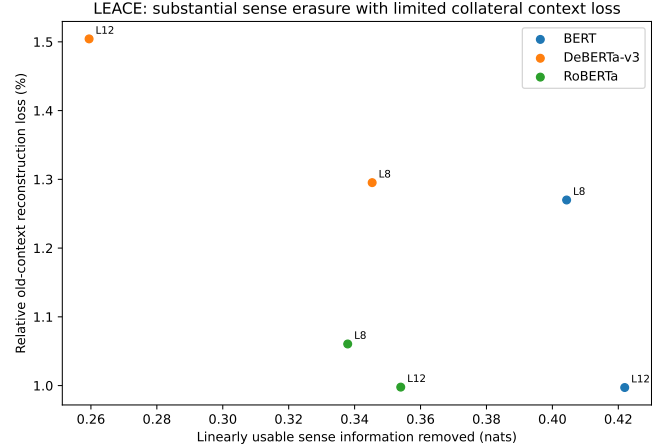


Figure 6: LEACE tradeoff averaged across base sizes. The x-axis is linearly usable sense information removed; the y-axis is relative loss of the fixed old-context reconstruction target.

different-sense AUC ranges from .764 to .912 (Table 4). These values provide external semantic validity for the geometry but do not test the controlled reorganization mechanism itself. Because multiple sentence pairs share lexical items, we do not use the naïve pair-level significance tests in the paper.

6 Discussion

6.1 Accessibility Is Not Geometric Simplicity

The main conceptual result is narrower than an uncertainty claim. Across the tested conditions, making lexical sense more linearly accessible does not impose a universal reduction in within-sense variation. This is compatible with a simple explanation: once *bank* is identified as financial, the representation can still encode mortgages, fraud, branches, locations, discourse roles, or syntax. Sense-related structure can become more explicit while other contextual degrees of freedom remain.

6.2 Why “Sense-Directed Reorganization”?

We use this phrase as a descriptive synthesis of four observations: controlled evidence differentiates senses; diagnostic updates contain a stronger supervised sense-subspace component; structural similarity to the pre-cue state changes; and old-context accessibility is not destroyed proportionately. None

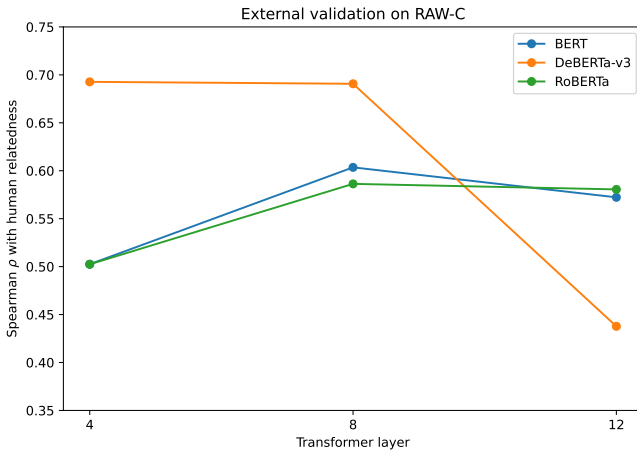


Figure 7: RAW-C descriptive alignment with graded human semantic relatedness. The layer dependence is substantial, particularly for DeBERTa-v3.

of these observations alone proves a unique computational mechanism. In particular, the supervised subspace and sense probe share label information, so the natural-context coupling should be read as evidence of *sense-structured change*, not as an unsupervised discovery that the model internally factorizes meaning into a privileged sense axis.

6.3 Architecture Differences Are Secondary

BERT, RoBERTa, and DeBERTa-v3 differ in raw within-sense trajectories, update alignment, and layer profiles, but the data do not justify stable labels such as “compressor”, “preserver”, or “enricher”. The more defensible result is the shared qualitative pattern plus architecture- and depth-dependent degrees of reorganization.

7 Limitations and Threats to Validity

No direct semantic-uncertainty estimate. We measure linear sense accessibility, not $H(S | C)$ or a calibrated posterior over human lexical uncertainty. Claims are therefore phrased in terms of accessibility and geometry.

Supervised subspace coupling. Both I_V and U_S use gold sense labels during training. A label-permuted or matched-random-subspace null would provide a stronger test that the observed coupling is specific to the true sense geometry. This control is not included in the current preprint.

Clustered natural and RAW-C observations. Natural transitions are nested within words, and RAW-C pairs are also grouped by lexical item. We therefore remove naïve pair-level significance claims. A submission-ready statistical pass should add lexical-item-cluster bootstrap/permutation intervals and multiplicity control.

Masking. The deletion robustness check is BERT-specific. It does not rule out architecture-specific masking artifacts for RoBERTa or DeBERTa-v3.

Old-context target. The TF-IDF/SVD target is a fixed lexical summary, not a complete measure of all contextual information. The SVD dimension and ridge regularization were fixed rather than tuned; the current experiment should therefore be interpreted comparatively. Future work could repeat the analysis with richer independently annotated context attributes.

Coarse lexical inventory. The controlled analyses cover 19 English nouns with coarse senses. RAW-C broadens lexical coverage but validates semantic alignment rather than the incremental intervention.

Linear erasure. LEACE concerns linear accessibility. Residual nonlinear sense information may remain, and a low held-out I_V does not imply that the model no longer contains or uses the concept in any nonlinear sense.

Representation-dependent geometry. Anisotropy and rogue dimensions affect cosine and Euclidean measures [Timkey and van Schijndel, 2021]. We avoid raw magnitude comparisons across architectures and treat standardized/whitened analyses as robustness diagnostics rather than coordinate-free truths.

Excluded exploratory PMI statistic. A post-hoc diagnostic of PMI in the local-context experiment mishandled unseen tokens. That statistic is excluded from all paper claims; the reveal order itself is label-independent and unaffected.

8 Conclusion

Across the tested contextual encoders and coarse ambiguous nouns, increased *linear accessibility* of lexical sense is not accompanied by a universal collapse of within-sense geometry. Controlled diagnostic evidence differentiates senses; nested diagnostic updates align more with supervised sense-discriminative directions while changing broader structure; a fixed old-context target remains largely recoverable; and substantial linear sense erasure causes only limited collateral context loss. Label-independent local context expansion and RAW-C provide complementary robustness checks. We therefore use **sense-directed representational reorganization** as a compact descriptive account of these observations: context can make lexical sense more accessible without making the contextual representation geometrically simple.

Reproducibility Statement

The repository contains executed notebooks for Experiments A–C, paper-facing summary tables, figure-generation code, and LaTeX source. Random seeds, model identifiers, lexical

sampling, and bootstrap settings are recorded in the notebooks. Large embedding caches and third-party datasets are not redistributed. The current public summary CSV exports are intended as paper-facing tables; the executed notebooks remain the provenance for the full analyses.

References

- Nora Belrose, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. LEACE: Perfect linear concept erasure in closed form. In *Advances in Neural Information Processing Systems*, volume 36, 2023. doi: 10.52202/075280-2884.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, pages 4171–4186, 2019. doi: 10.18653/v1/N19-1423.
- Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. Amnesic probing: Behavioral explanation with amnesic counterfactuals. *Transactions of the Association for Computational Linguistics*, 9:160–175, 2021. doi: 10.1162/tacl.a.00359.
- Kawin Ethayarajh. How contextual are contextualized word representations? comparing the geometry of BERT, ELMo, and GPT-2 embeddings. In *Proceedings of EMNLP-IJCNLP*, pages 55–65, 2019. doi: 10.18653/v1/D19-1006.
- Pengcheng He, Jianfeng Gao, and Weizhu Chen. DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. *arXiv preprint arXiv:2111.09543*, 2021.
- John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Proceedings of EMNLP-IJCNLP*, 2019.
- Zhimin Hu, Lanhao Niu, and Sashank Varma. Language models represent and transform concepts with shared geometry. *arXiv preprint arXiv:2607.04525*, 2026.
- Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *Proceedings of ICML*, pages 3519–3529, 2019.
- Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. RoBERTa: A robustly optimized BERT pretraining approach. *arXiv preprint arXiv:1907.11692*, 2019.
- Daniel Loureiro, Kiamehr Rezaee, Mohammad Taher Pilehvar, and Jose Camacho-Collados. Analysis and evaluation of language models for word sense disambiguation. *Computational Linguistics*, 47(2):387–443, 2021. doi: 10.1162/coli.a.00405.
- Vishal Pandey and Gopal Singh. Trajectory geometry of transformer representations across layers. *arXiv preprint arXiv:2606.09287*, 2026.
- Tiago Pimentel, Rowan Hall Maudslay, Damian Blasi, and Ryan Cotterell. Speakers fill lexical semantic gaps with context. In *Proceedings of EMNLP*, pages 4004–4015, 2020. doi: 10.18653/v1/2020.emnlp-main.328.
- William Timkey and Marten van Schijndel. All bark and no bite: Rogue dimensions in transformer language models obscure representational quality. In *Proceedings of EMNLP*, pages 4527–4546, 2021. doi: 10.18653/v1/2021.emnlp-main.372.
- Sean Trott and Benjamin Bergen. RAW-C: Relatedness of ambiguous words in context (a new lexical resource for english). In *Proceedings of ACL-IJCNLP*, pages 7077–7087, 2021. doi: 10.18653/v1/2021.acl-long.550.
- Yile Wang and Yue Zhang. Lost in context? on the sense-wise variance of contextualized word embeddings. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 2023. doi: 10.1109/TASLP.2023.3337643.
- David Yenicelek, Florian Schmidt, and Yannic Kilcher. How does BERT capture semantics? a closer look at polysemous words. In *Proceedings of the Third BlackboxNLP Workshop*, pages 156–162, 2020. doi: 10.18653/v1/2020.blackboxnlp-1.15.

A Selected Quantitative Tables

A.1 Controlled final-layer geometry

Model	k	ΔAcc	ΔW	ΔB
BERT	1	.189	.0006	.0506
BERT	2	.203	-.0117	.0876
BERT	4	.123	-.0211	.0877
RoBERTa	1	.166	.0011	.0038
RoBERTa	2	.194	.0000	.0074
RoBERTa	4	.155	.0003	.0093
DeBERTa-v3	1	.043	.0078	.0054
DeBERTa-v3	2	.047	.0139	.0096
DeBERTa-v3	4	.068	.0144	.0112

Table 2: Representative layer-12 controlled effects. Values are diagnostic minus the five-seed random mean. These raw-space magnitudes are not used for cross-model strength comparisons.

A.2 LEACE absolute reconstruction levels

A.3 RAW-C external validation

Model	L	Ctx pre	Ctx fresh	Ctx frozen	Rel. loss
BERT	8	.1785	.1762	.1762	1.27%
BERT	12	.1802	.1784	.1785	1.00%
RoBERTa	8	.1754	.1736	.1736	1.06%
RoBERTa	12	.1867	.1848	.1848	1.00%
DeBERTa	8	.1821	.1798	.1799	1.30%
DeBERTa	12	.1572	.1549	.1548	1.50%

Table 3: Old-context reconstruction cosine before LEACE and after LEACE with fresh/frozen decoders, averaged over base sizes.

Model	Layer	ρ	AUC
BERT	4	.503	.818
BERT	8	.604	.859
BERT	12	.572	.842
RoBERTa	4	.502	.804
RoBERTa	8	.586	.850
RoBERTa	12	.581	.847
DeBERTa-v3	4	.693	.877
DeBERTa-v3	8	.691	.912
DeBERTa-v3	12	.438	.764

Table 4: RAW-C effect sizes. We report them descriptively because sentence pairs are clustered by lexical item; pair-level p -values are not used for inference.